Apartment Complex Management System

Milestone 1 – User Requirement Analysis

1. Objective

To streamline apartment society operations through a centralized platform for maintenance payments, complaint management, facility booking, visitor management, emergency assistance, and society communication. The system improves transparency, accountability, and operational efficiency for residents and society officials.

2. Identification of Users

Primary Users	Secondary Users	Tertiary Users
Residential Members / Flat Owners	Non-residential Flat Owners	Security Guards
Society Officials	Visitors / Guests	Cleaning Staff
		Maintenance Staff
		Service Providers

3. User Descriptions

- **Residential Members / Flat Owners:** Use the platform to pay maintenance fees, raise complaints, book common facilities, generate guest passes, receive announcements, and access society services.
- **Society Officials:** Manage residents, payments, complaints, facility bookings, announcements, and society operations.
- **Non-residential Flat Owners:** Use the platform to remotely view announcements, track payments and complaints, and generate guest passes.
- **Visitors / Guests:** Receive digital guest passes generated by residents for secure society entry.
- **Security Guards:** Carry out visitor verification and security duties based on information provided by society officials.
- **Cleaning Staff:** Perform cleaning tasks coordinated by society officials.
- **Maintenance Staff:** Resolve maintenance issues assigned by society officials.
- **Service Providers:** Provide maintenance services when required.

4. User Research

Our team interviewed residential members, society officials, and non-residential flat owners to understand current challenges. The interviews focused on identifying pain points rather than suggesting solutions, and the findings were converted into Agile user stories.

5. Key Pain Points Identified

- Manual maintenance payment tracking.
- Lack of payment confirmation.
- Complaints managed through WhatsApp.
- No complaint status visibility.
- Manual facility booking causes conflicts.
- Society information scattered across chats.
- No centralized management platform.
- No quick method to notify security during emergencies.

6. User Stories

Epic 1: Maintenance Management

Story 1.1: Maintenance Payment Confirmation

As a residential member/flat owner, I want to receive confirmation after making a maintenance payment, so that I know my payment has been successfully recorded.

Story 1.2: Maintenance Payment Reminders

As a residential member/flat owner, I want to receive reminders for maintenance payments, so that I do not miss payment deadlines.

Story 1.3: Maintenance Record Management

As a society official, I want to maintain accurate maintenance payment records, so that I can avoid sending incorrect payment reminders.

Epic 2: Complaint Management

Story 2.1: Complaint Status Tracking

As a residential member/flat owner, I want to track the status of my complaints, so that I know whether my reported issues are being addressed.

Story 2.2: Centralized Complaint Handling

As a society official, I want to receive and track complaints through a centralized system, so that resident issues do not get overlooked.

Story 2.3: Complaint Prioritization

As a society official, I want to prioritize complaints based on urgency, so that critical issues are addressed first.

Epic 3: Facility Management

Story 3.1: Booking Confirmation

As a residential member/flat owner, I want to receive confirmation for facility bookings, so that I can be confident that my reservation is secured.

Story 3.2: Shared Facility Booking

As a residential member/flat owner, I want to book time slots for shared facilities, so that I can avoid overcrowding.

Story 3.3: Booking Schedule Management

As a society official, I want to manage facility bookings through a shared schedule, so that double bookings do not occur.

Epic 4: Visitor & Emergency Management

Story 4.1: Digital Guest Pass

As a residential member/flat owner, I want to generate digital guest passes, so that my guests can enter the society without delays.

Story 4.2: Emergency SOS

As a residential member/flat owner, I want an SOS button to alert security during emergencies, so that I can receive immediate assistance.

Epic 5: Society Administration

Story 5.1: Resident Dashboard

As a residential member/flat owner, I want to view my payments, complaints, and bookings in one place, so that I can easily track my interactions with the society.

Story 5.2: Centralized Records

As a society official, I want to access society records and communications from one place, so that I can quickly find relevant information.

Epic 6: Non-residential Owner Services

Story 6.1: Society Updates

As a non-residential flat owner, I want to view important society announcements in one place, so that I do not miss important updates.

Story 6.2: Property Status

As a non-residential flat owner, I want to view my maintenance payments and complaint status, so that I can manage my property remotely.

Story 6.3: Visitor Access

As a non-residential flat owner, I want to generate digital guest passes for visitors, so that they can access my property without manual coordination.

7. Interview Evidence

Interview recordings and transcripts have been uploaded to the designated Google Drive folder here.